

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front parallelogram is blue and the back one is a light green color. Both are oriented diagonally from the top-left towards the bottom-right.

Quantum safe federated learning framework for cyber threat detection in 5G-enabled IOT ecosystems



Introduction

5G-enabled IoT ecosystems generate massive amounts of sensitive network traffic data and are increasingly vulnerable to sophisticated cyber threats such as: ICMP flood attacks, SYN flood attacks, Port scanning, UDP flood attack, etc.

Traditional centralized deep learning approaches can achieve high accuracy, but they suffer from several problems:

- lack of privacy preservation
- scalability issues
- centralized data sharing risks
- vulnerability to parameter leakage

To address these limitations, the paper proposes a quantum-safe federated learning framework called **DeepFL-CTI**.



Problem Statement addressed

- Cyber Threats targeting 5G IOT ecosystems (ICMP flood, SYN flood, port scans, etc)
- Cyber Threats targeting FL (model parameter leakage, insecure parameter sharing, privacy leakage during aggregation, etc)



Key Contributions of the Paper

1. **Hybrid MobileNet-V1 + BiGRU Model:** The paper proposes a hybrid deep learning architecture combining MobileNet-V1 and Bidirectional GRU (BiGRU) for efficient cyber-threat detection in 5G IoT environments.
2. **Federated Learning Framework:** The framework enables: collaborative learning, decentralized training and privacy-preserving model updates without sharing raw client data.
3. **Post-Quantum Secure Aggregation:** The framework integrates: lattice-based cryptography Learning With Errors (LWE) to provide quantum-safe secure communication between clients and server.
4. **Evaluation on 5G-NIDD Dataset:** The proposed framework is evaluated using multiple metrics: accuracy, precision, recall, F1-score, TPR, FPR, FNR, NPV and outperforms previous related works.



System Model

1. **Trusted Authority (TA):** Responsible for -
 - generating cryptographic keys
 - establishing secure communication
2. **IoT Clients:** Responsible for -
 - local training on private IoT data
 - encrypting model updates
3. **Cloud Server:** Responsible for -
 - aggregating encrypted model updates
 - updating the global model



Workflow

Step 1: Dataset Preparation: The 5G-NIDD dataset is: cleaned, preprocessed, normalized encoded, etc. Then divided across multiple IoT clients. Each client receives its own local private dataset.

Step 2: Key Generation (Trusted Authority): The Trusted Authority generates public key and private key using the **LWE Cryptographic Scheme**.

$$\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$$

Where:

- $\mathbf{A} \rightarrow$ random matrix
- $\mathbf{s} \rightarrow$ secret/private vector
- $\mathbf{e} \rightarrow$ small error vector
- $\mathbf{b} \rightarrow$ public component



Workflow

Step 3: Distribution of Initial Global Model: The cloud server initializes MobileNet-V1 + BiGRU global model. The server sends initial model weights and public encryption key to all IoT clients.

Step 4: Secure Aggregation at Server:

Each IoT client:

1. receives global model weights
2. trains locally on its private data
3. updates model parameters

Using MobileNet + BiGRU model



Workflow

Step 5: Encryption of Local Model Updates : After local training each client encrypts updated model weights using LWE Encryption. This ensures privacy preservation and quantum-safe communication. So client never sends raw data only encrypted data to the server.

Step 6: Secure Aggregation at Server:

The cloud server performs: Federated Averaging (FedAvg) on encrypted model parameters. The server:

- aggregates updates from all clients
- creates updated global model weights

WITHOUT accessing any raw data.



Workflow

Step 7: Decryption of Aggregated Model: Using the private key generated by Trusted Authority, the aggregated global model is decrypted to obtain updated global parameters.

Step 8: Global Model Update: The server updates Global DeepFL-CTI Model with newly aggregated parameters.

Step 9: Redistribution to Clients: Updated global weights are again sent to all clients. The FL cycle repeats for multiple communication rounds until convergence is achieved.



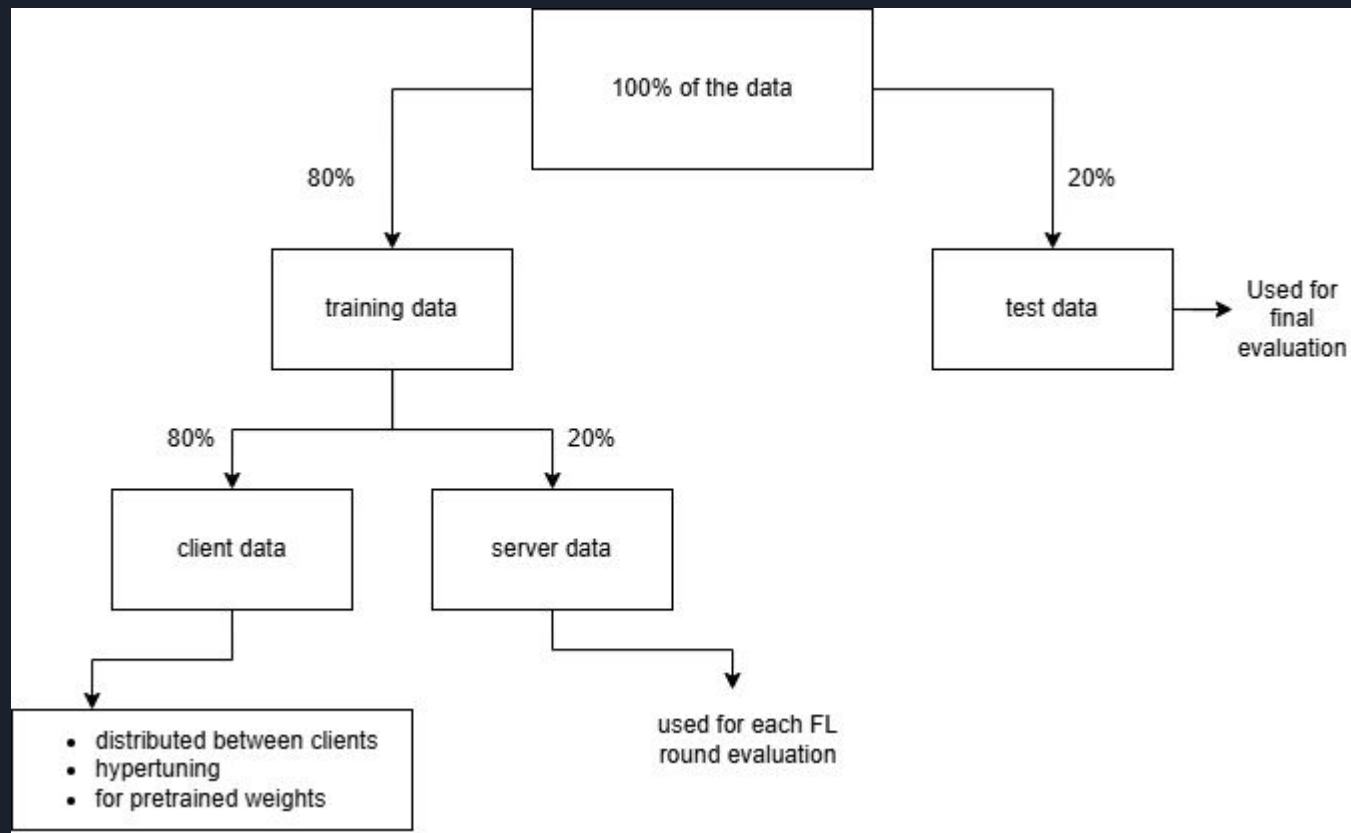
Final Evaluation

After convergence the final global model performs Cyber-Threat Detection on 5G IoT network traffic

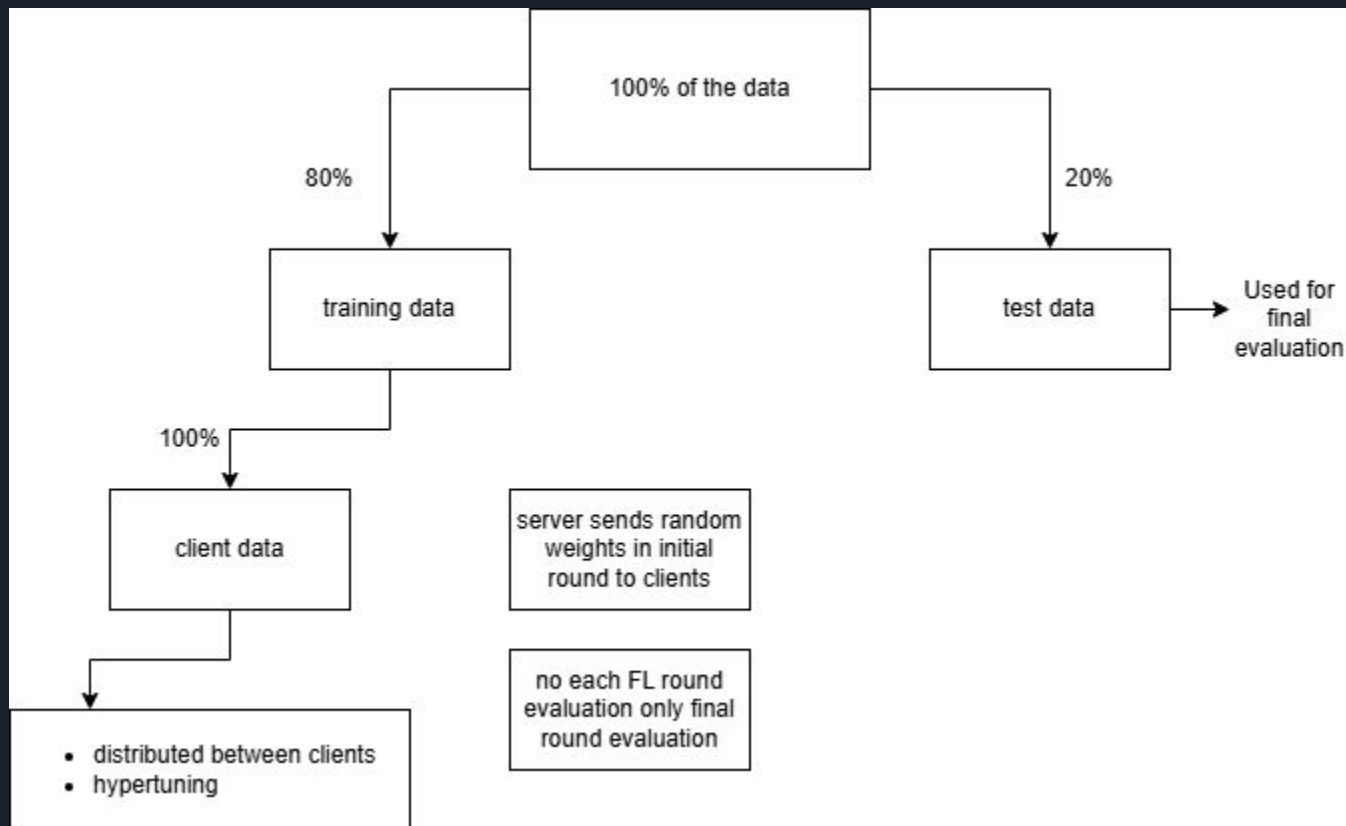
Detecting:

- ICMP flood
- UDP flood
- SYN scan
- DDoS
- port scan
- other attacks

My approach 1



My approach 2





Results & Discussion

The best results came from approach 1 after implementing DeepFL-CTI model for 13 rounds for $k = 5$.

Accuracy = **97.77%**

F1 Score = **0.9772**

Macro Avg = **0.90**

Precision = **0.9777**



Approach 1 Results Comparison

K value	AC(a)	PC(a)	F1(a)	AC(b)	PC(b)	F1(b)
3	99.80%	0.9962	0.9968	95.44%	0.9557	0.9544
5	99.82%	0.9953	0.9957	97.77%	0.9777	0.9772
7	99.73%	0.9959	0.9956	95.34%	0.9561	0.9537
9	99.75%	0.9965	0.9946	95.55%	0.9572	0.9555

a = Paper results, b = My approach 1 results

AC = Accuracy,

PC = Precision,

F1 = F1 Score



Approach 2 Results Comparison

K value	AC(a)	PC(a)	F1(a)	AC(b)	PC(b)	F1(b)
3	99.80%	0.9962	0.9968	97.26%	0.9737	0.9729
5	99.82%	0.9953	0.9957	94.43%	0.9451	0.9445
7	99.73%	0.9959	0.9956	93.78%	0.9419	0.9382
9	99.75%	0.9965	0.9946	92.88%	0.9322	0.9291

a = Paper results, b = My approach 2 results

AC = Accuracy,

PC = Precision,

F1 = F1 Score



Reasons for 2-4% difference of results

- **Client Data Distribution Differences:** Federated Learning is very sensitive to Non-IID client partitioning. Small changes in attack distribution, class balance and client heterogeneity can easily change accuracy by small %.
- **Encryption Overhead:** Integrating: LWE encryption and secure aggregation can slightly affect:
 - convergence
 - numerical precision
 - aggregation stability

Especially if quantization/scaling is used.



Reasons for 2-4% difference of results

- **Fewer Extensive search and ideal scenario:** Paper likely works in optimized convergence, stronger local training and ideal setup. Paper likely used extensive tuning, many experiments, manually optimized setup.
- **Statistical Variance in FL:** Federated Learning is inherently:
 - stochastic
 - unstable
 - client-sensitive
- **Random Client Selection:** 'random.sample()' This changes: participating clients, local gradients and aggregation path.



Limitations and Future Scope

- **Limited Real-Time Deployment Validation:**

The framework is tested mainly on benchmark datasets and simulated FL settings

But not on:

- actual live 5G IoT infrastructure
- production-grade deployments

So real-world latency and scalability remain uncertain.

- **Integration with Blockchain:** Blockchain can be integrated for -

- decentralized trust
- tamper-proof aggregation
- secure client authentication

This is a very strong future scope point.



Limitations and Future Scope

- **Transformer-Based Federated Models:**

Future work may explore: Transformers, Vision Transformers, Graph Neural Networks, Attention-based IDS, etc for better feature learning.

- **Explainable AI (XAI) Integration:** Cybersecurity systems require interpretability.

Future systems can integrate:

- SHAP
- LIME
- attention visualization

to explain attack predictions.



Limitations and Future Scope

- **Limited Analysis of Extreme Non-IID Scenarios:**

Real-world IoT systems may contain:

- highly skewed attack classes
- clients with very little data
- completely different traffic patterns

which may reduce convergence.

- **No Robust Defense Against Malicious Clients:**

Paper does not deeply address:

- poisoning attacks
- Byzantine attacks
- malicious model updates



Limitations and Future Scope

- **Dataset Dependency:**

The model is evaluated mainly on 5G-NIDD dataset

Performance on: other IoT datasets, unseen attacks, zero-day attacks
is not extensively validated.

- **Lightweight Encryption Schemes:** LWE is secure but computationally expensive.

Future work may explore:

- optimized lattice cryptography
- lightweight post-quantum schemes
- hybrid encryption

for better efficiency.



THANK YOU